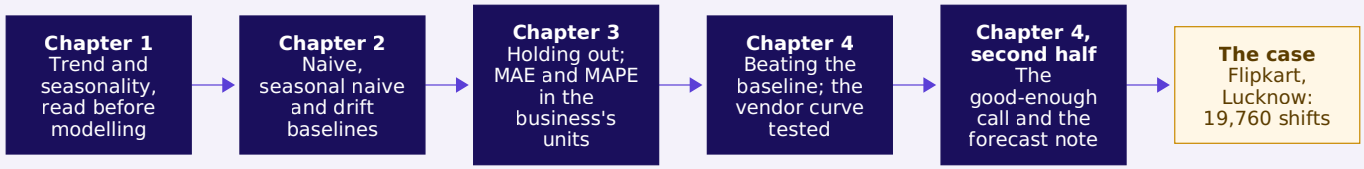
FORECASTING BASICS

how much does the plan deliver, and how wrong could it be?

Week 4 Friday. Five topics on a chai stall at a hostel gate (invented), then Flipkart's Lucknow fulfilment centre, where the planner books packer shifts for the eight weeks beginning 12 October to 30 November 2026 between a vendor's plus 22 percent and finance's plus 4. Illustrative case: the company is real, the situation and the data are synthetic, built for this class.

C2_W04_D05_forecasting_basics_STUDENT
Verified 13 Oct 2026
7 panels, 2 diagrams

1 How do the five topics connect, and where does the case use them?



Crux. Read the shape, copy it, score the copy on periods it never saw, keep only what beats the copy, and stop when nothing better would change the order.

2 Trend and seasonality, read before modelling: what does a series tell?

- A **season** repeats on a fixed beat, up or down; its **length** is the periods before it repeats: 7 days at the invented stall, 52 weeks at Flipkart.
- Read growth at the same point a season apart: a week apart shows a week's change; Monday 110 against 100 a year earlier is 10 percent a year; Monday over Sunday, 100 over 60, is the beat.
- At the stall (invented) the exam fortnight runs about 1.4 times an ordinary one, Diwali week about 0.4; a yearly copy's misses have a typical size of about 4.7 cups.
- `y = T + S + R` when the seasons keep their size; `y = T x S x R` when they grow with the level.

Crux. Name the beat and its length before any rule; at Lucknow, about 4 percent a year and a festive peak.

3 Naive, seasonal naive and drift baselines: when does a copy fail?

Rule	Copies	The invented stall's next Sunday	At Lucknow
Naive	the last value, `y[T]`	103	the week of 22 June
Seasonal naive	one season back, `y[T+h-m]`, h up to m	61, against 60 sold	the house bot, m = 52
Drift	the line extended, `y[T] + h x (y[T] - y[1]) / (T - 1)`	about 104.7	a step of 70 a week
Festival-aligned	the same distance from last year's festival	Diwali's worst miss 13, not 71	this programme's extension

Crux. On a strong beat the plainest copy that respects it wins; a calendar copy fails when the festival moves.

4 Holding out; MAE and MAPE in the business's units: how is a forecast scored?

- Hide the latest periods, at least as many as the horizon: the stall's (invented) days 12 to 14 from day 11; Lucknow's 13 weeks from 22 June 2026, the plan reaching ten weeks past the data.
- MAE averages the misses' sizes, in the unit the cost is in: the stall's 5.0 cups; MAPE in percent, 4.8, explodes near zero and pulls a rule low.
- A miss here is forecast minus sold; the textbook writes sold minus forecast, so its signs flip.
- Keep the **lean**, the signed average: -1.7 cups, running low. Price it: Rs 150 at Rs 12 short and Rs 6 unsold.
- Same MAE, different rupees: +10, +10 costs Rs 120; +10, -10 costs Rs 180.

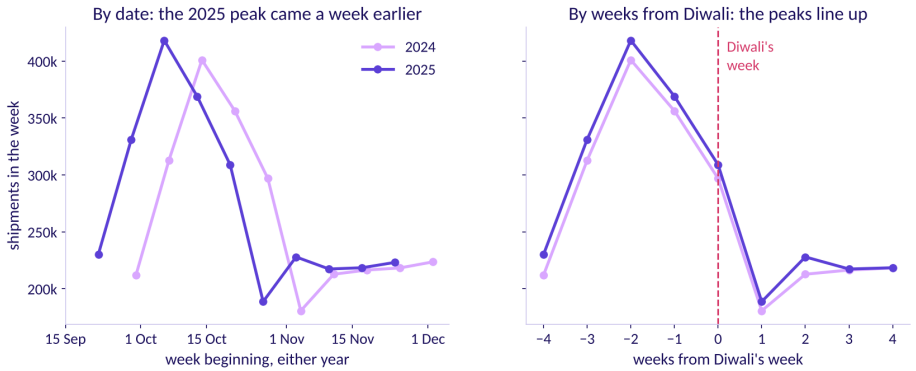
Crux. Fix the measure before scoring, in the business's unit, and report the size and the direction together.

5 Beating the baseline and the good-enough call: when does a forecast earn trust?

- Fit is closeness on seen periods: the stall's (invented) curve fits four days with an error of 0 and misses day 5 by 66.
- The nephew's mean (invented) scores 12.4 cups on hidden days against the seasonal naive's 4.0: it earns no place.
- Good enough is a claim about the decision: improve only while the change is worth the work; the seasonal naive already orders the cans a perfect forecast would.
- The note: the number, the method, the error bar in the unit and in rupees, what would make it wrong.

Crux. Nothing goes to the order unless it beats the seasonal naive on hidden periods by enough to pay for itself.

6 The case: where does the peak fall, and how many shifts?



On 13 hidden weeks	MAE	Misses priced
Festival-aligned	6,615	Rs 13.5 lakh
Seasonal naive, the house bot	9,831	Rs 30.5 lakh
Vendor's curve, fitted within 1.4 percent	16,400	Rs 11.9 lakh, cheapest only at Rs 25 a late shipment

The plan books festival-aligned's forecast over 120 a shift, rounded up: 19,760 shifts, 3,625 in the week of 19 October. Error bar: a typical miss of 6,615 shipments a week, about Rs 35,800 if high, Rs 1.65 lakh if low; at the peak the hidden weeks' range means 3,413 to 3,906 shifts. On 2025's festival, with no growth, it missed by 10,575. Beyond the row: the plan quantile, planning to cover 0.82 of weeks, raises it to 20,275.

Crux. Copy by distance from Diwali, score on hidden weeks, price both directions, book in whole shifts.

7 What does the interviewer listen for?

[S] "How do you evaluate a forecast?" Hold out the latest weeks, as many as the horizon; score in shipments and rupees against the seasonal naive; back-test a festival, since the hidden weeks hold none. [F] "What baseline must any forecast beat, and why?" Naive with no season, the seasonal naive with one, plus growth when it trends: a free copy. [F] "A vendor promises 22 percent growth; how do you test it?" Read the 22 against the same weeks, score its curve on weeks it never saw, and ask for its festival record. Tags: [S] asked everywhere, [F] frequent in GCC and product screens.

Crux. Name the hidden weeks, the unit, the copy to beat and the direction that costs more.

THE WORDS ON THIS SHEET

Festival-aligned This programme's extension: a copy at the same distance from last year's...

MAE, MAPE, lean The average miss's size; as a percent; with signs kept, which an...

Error bar The typical miss on unseen periods, priced both ways, and the range of...